



What Makes Something Alive?

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The 6 Characteristics of Life

An object is alive only if it does **all six**:

- Made of **cells**
- Use **energy**
- **Grow & develop**
- **Reproduce**
- **Respond** to surroundings (stimulus → response)
- Maintain **homeostasis** (stable inside)

Key Terms

Organism – one complete living thing • **Homeostasis** – keeping a stable internal environment • **Stimulus** – a change that causes a reaction • **Response** – what the organism does back.

⚠ **Movement ≠ life.** A river moves and a flame "grows," but neither is alive. Use the whole checklist.

Classification & Its History

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Classification = sorting living things into groups by shared traits. **Taxonomy** = the science of naming & classifying. The more traits two organisms share, the more closely related they usually are.

When / Who	What they did
~350 BCE — Aristotle	First sorted life into just plants & animals
1735 — Carolus Linnaeus	Built today's system; gave each species a two-word name
1600s — microscopes	Found tiny organisms that didn't fit "plant" or "animal" — new kingdoms added
Today — DNA & cells	Group by genes, not just looks — far more accurate

🔑 The system keeps improving as our **tools** improve (eye → microscope → DNA).

The 8 Levels of Classification

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Broadest → narrowest. As you move down, groups get smaller and organisms more alike:

Domain › Kingdom › Phylum › Class › Order › Family ›
Genus › Species

★ **Memory trick:** Dear King Philip Came Over For Good Soup.

One gray wolf, all the way down

Level	Group
Kingdom	Animalia
Phylum	Chordata (backbone)
Class	Mammalia
Order	Carnivora
Family	Canidae (dog family)
Genus	<i>Canis</i>
Species	<i>Canis lupus</i>

The Six Kingdoms of Life

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Scientists sort all life by 3 questions: **one cell or many? nucleus or not? how does it get food?**

Kingdom	Cell Type	# Cells	Gets Food By...
Archaeobacteria	Prokaryote (no nucleus)	One	Making or absorbing — lives in extremes
Eubacteria	Prokaryote (no nucleus)	One	Absorbing — common everywhere
Protists	Eukaryote (nucleus)	One or many	Varies (amoeba, algae)
Fungi	Eukaryote	Mostly many	Absorbing (mushroom, mold, yeast)
Plants	Eukaryote	Many	Making food (photosynthesis)
Animals	Eukaryote	Many	Eating other organisms

🔑 **The big split:** Archaea & Bacteria have **no nucleus** (prokaryotes). The other four keep DNA in a nucleus (eukaryotes).

Dichotomous Key

A step-by-step list of **either/or** questions ("divided into two"). Each answer splits the possibilities in half until one organism is left.

- Always start at step 1
- Read **both** choices first
- Keep going until you reach a **name**, not another question

Binomial Nomenclature

Every species gets a **two-word** scientific name (Genus + species):

- **Genus** first, **Capitalized**
- **species** second, lowercase
- Whole name *italicized* (or underlined by hand)

e.g. *Canis lupus*, *Homo sapiens*.

 Never *Homo Sapiens* — only the genus is capitalized.

Quick Check — Kingdom or Characteristic?

A. Name the kingdom:

1. Mushroom, made of many cells, absorbs food → ____
2. Oak tree, makes its own food → ____
3. Bacteria living in a boiling hot spring → ____

B. Write the genus & species correctly:

1. house cat: felis catus → _____
2. Order these from broad → narrow: Family, Domain, Genus, Class → _____

I Can... (Self-Check)

- | | |
|--|--|
| ☐ List the 6 characteristics of all living things | ☐ Use a dichotomous key to identify an organism |
| ☐ Explain why scientists classify life | ☐ Write a scientific name with correct capitalization |
| ☐ Order the 8 levels (Domain → Species) | ☐ Describe how the system improved with new tools |
| ☐ Tell the six kingdoms apart by cell & food | ☐ Tell prokaryotes from eukaryotes |